

Some review questions - 10th week

Q1 Using the geometric interpretation of the determinant, justify the following identity for some $a, b, c, d \in \mathbb{R}$:

$$\begin{vmatrix} 2a & 2b \\ c & d \end{vmatrix} = 2 \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

Remark: Draw parallelograms and explain the formula using the relations between areas.

Q2 What can be said about the determinant of a matrix that has a row or a column consisting entirely of zeros? Justify your answer for 3×3 matrices using the geometric interpretation of volume.

Q3 Prove that if a matrix $A \in \mathbb{R}^{n \times n}$ has a row or a column consisting entirely of zeros, then $\det(A) = 0$. You may use the linearity of the determinant with respect to rows and columns.

Q4 Given $a_1, b_1, a_2, b_2, c, d \in \mathbb{R}$, justify, using the geometric interpretation of the determinant, that:

$$\begin{vmatrix} a_1 + a_2 & b_1 + b_2 \\ c & d \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ c & d \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ c & d \end{vmatrix}.$$

Remark: Draw parallelograms and explain the formula using the relations between areas.

Q5 Given two matrices $A, B \in \mathbb{R}^{n \times n}$, and assuming A is invertible whenever A^{-1} appears, determine whether each of the following identities is true or false (no justification needed):

$$\begin{aligned} \det(A + B) &= \det(A) + \det(B), & \det(AB) &= \det(A) \det(B), \\ \det(A^\top) &= \det(A), & \det(A^{-1}) &= \det(A). \end{aligned}$$

Q6 By expanding the following determinant along the second row, provide an expression involving three minors of size 2×2 :

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

Q7 Given $h \in \mathbb{R}$ and $B \in \mathbb{R}^{2 \times 2}$, give a closed-form expression for $\det(A)$ depending on h and B when A is the

block matrix

$$\begin{bmatrix} h & 0 \\ 0 & B \end{bmatrix} \in \mathbb{R}^{3 \times 3}.$$

Justify this formula using the geometric interpretation of volume.

- Q8** Give the recursive definition of the determinant (do not forget to define the determinant of a 1×1 matrix). Given $i, j \in \{1, \dots, n\}$, you may use the notation $M_{ij} \in \mathbb{R}^{(n-1) \times (n-1)}$ for the matrix obtained by deleting the i^{th} row and the j^{th} column of A .

- Q9** Given a matrix $A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$, provide the Laplace expansion formula for the determinant of A along the column $k \in \{1, \dots, n\}$. Given $i, j \in \{1, \dots, n\}$, you may use the notation $M_{ij} \in \mathbb{R}^{(n-1) \times (n-1)}$ for the matrix obtained by deleting the i^{th} row and the j^{th} column of A .

Q10 Compute

$$D = \begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}.$$

Q11 Consider $i, j \in \{1, \dots, n\}$, $\beta \in \mathbb{R}$ and $\alpha \in \mathbb{R}$, $\alpha \neq 0$. Given a matrix $A \in \mathbb{R}^{n \times n}$, suppose that, after the row operations $R_i \leftrightarrow R_j$, $R_i \rightarrow \alpha R_i$ and $R_i \rightarrow R_i + \beta R_j$, we respectively obtain the matrices B, C, D . Give three formulas relating $\det(B)$, $\det(C)$, $\det(D)$ and $\det(A)$.

Q12 Compute

$$D = \begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 4 \end{vmatrix}.$$

Q13 Provide a formula for the determinant of diagonal and triangular matrices.

Q14 What can be said about A if $\det(A) \neq 0$? (only state one key property.)

Q15 Prove that $\det(A) \neq 0$ if and only if A is invertible. In your proof you should make use of the following tools/facts:

- Gauss-Jordan elimination,
- how an elementary row operation modifies the determinant of a matrix,
- the expression of the determinant of triangular matrices.

Q16 Give the definition of a linear transformation.

Q17 Given a matrix $A \in \mathbb{R}^{m \times n}$, show that the mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(x) = Ax$ is a linear transformation. Deduce that the mapping $S : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $S((x, y)) = x - y$ is a linear transformation.

Q18 Given a vector $v \in \mathbb{R}^n$, show that the mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $T(x) = \langle x, v \rangle$ is a linear transformation.

Q19 Is the mapping $T : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T((x, y)) = \sqrt{x^2 + y^2}$ a linear transformation? Prove your result.

Q20 Give three rules to check graphically whether a mapping is a linear transformation.

Q21 Give a general method, using the standard basis of $\mathbb{R}^n$, to determine the matrix associated with a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$.